Mission Definition and Requirements for a Conceptual Reusable Launch Vehicle (CRLV-1)

A Sustainable, Reusability-Driven Design Framework — v1.0 (refined)

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Abstract.

This paper presents the mission definition and Level-0 / Level-1 requirements for CRLV-1, a conceptual partially reusable launch vehicle targeting 1,200–2,000 kg of payload to low Earth orbit (LEO) or sun-synchronous orbit (SSO). The work draws on a structured review of 2025–2026 reusable launch vehicle (RLV) developments, with particular attention to the rapid emergence of Chinese state and commercial programmes. A defining feature of the framework is the explicit, **unit-consistent** definition of a Green Reusability Index (GRI) and a formal requirement to evaluate sea-based net-capture recovery as an alternative to conventional propulsive landing. Where public 2025–2026 sources disagree on the precise designation of the Chinese booster that demonstrated net-capture in 2026 (variously described as Long March 10, 10B, or 12A), the document uses the conservative family label *Long March 10/12A* and flags the naming uncertainty. All quantitative claims are either cross-referenced to a 2025–2026 source or explicitly labelled as *estimate* or *illustrative*. A self-critical review identified twelve issues in the original draft; all twelve are addressed in this refined v1.0. Fourteen Level-1 requirements, six Level-0 objectives, and a unit-consistent GRI form the basis for the Day 2–10 trade studies.

■■■■■■	■■ / ■■	v1.0 ■■■■
■■■■ (L0-01)	1,200 / 2,000 kg (LEO 500 km / SSO 700 km)	
■■■■■ (L0-02)	10 / 20 ■	—
■■ (L0-03)	< $3,500 / < $2,500 per kg	v0 $2,800 → v1 $3,500 (■■■■■■■■)
■■■■ (L0-04)	< 15 t CO■e/t payload	GRI ■■■■■■ (kg payload / kg CO■e)
■■■ (L0-05)	30 / 14 ■	—
■■■ (L0-06)	■■ ≥ 0.95 (10 ■)	■■ ≥ 0.994 / 0.997

Table 1. CRLV-1 Day-1 executive summary (v1.0). Pink-highlighted cells = v0 → v1.0 changes.

1. Introduction

The rapid maturation of reusable launch vehicle (RLV) technology, led by SpaceX's Falcon 9 and accelerated by Chinese state and commercial programmes in 2025–2026, has fundamentally changed the economics of access to space. As of mid-2026, multiple Chinese vehicles have either achieved or attempted first-stage recovery, with a publicly reported demonstration of a sea-based **net-capture** architecture on a Chinese methalox booster representing a noteworthy first for the Chinese orbital-class programme *(Ars Technica, 2026)*. Public sources disagree on the precise designation of that vehicle, variously referring to it as *Long March 10*, *Long March 10B*, or *Long March 12A*; the present document uses the conservative family label *Long March 10/12A* throughout and flags the naming uncertainty explicitly in the references and risk register (R08).

This paper documents the Day-1 mission-definition process for **CRLV-1** (Conceptual Reusable Launch Vehicle 1), an academic, demonstrator-class RLV positioned in the payload gap between small dedicated launchers (e.g., Rocket Lab Electron, ~300 kg) and medium-lift reusable vehicles (e.g., Falcon 9, ~22.8 t). The framework integrates performance, reusability, cost, and sustainability considerations; introduces the GRI as a primary figure of merit; and explicitly retains a propulsive-vs-net-capture recovery trade for the Day-6 deep-dive.

2. Research Methodology and Data Sources

A systematic literature and news review was conducted on 2025–2026 RLV developments. Primary public sources included SpaceNews, Ars Technica, Global Times, China-in-Space, manufacturer press releases, and market analyses. Quantitative claims were cross-verified against at least two independent sources wherever possible; values that could not be cross-verified are explicitly flagged as *estimate*.

Vehicle	Operator	LEO payload (reusable, kg)	Status / note
Falcon 9 Block 5	SpaceX	22,800	Operational; >20 flights per booster; propulsive landing.
New Glenn	Blue Origin	45,000	Maiden flight 2025; methalox; 7 m fairing.

Vehicle	Operator	LEO payload (reusable, kg)	Status / note
Zhuque-3	Landspace (CN)	~18,300	Stainless-steel methalox; maiden orbital flight Dec 2025 (recovery attempt failed); reco
Long March 10/12A*	CASC (CN)	~12,000 (est.)	*Designation uncertain in public sources; sea-based net-capture demonstrated 2026.
Hyperbola-3	iSpace (CN)	8,500	Kerolox; maiden flight slipped to 2026; dedicated drone-ship procured.
Pallas-1	Galactic Energy (CN)	8,000	Kerolox; assembly and static fires 2025; first flight 2026.
CRLV-1 (proposed)	This work	1,200-2,000	Conceptual demonstrator; methalox; recovery TBD (propulsive or net).

Table 2. Reference vehicles analysed (reusable configuration unless noted).

Sustainability data were drawn from recent life-cycle assessment (LCA) studies of launch vehicles, which consistently indicate that traditional RP-1/LOX systems produce on the order of 15–20 t CO_2e per tonne of payload delivered to LEO. The mid-point value of ~19 t CO_2e/t used elsewhere in this document is illustrative within that published range, not a precise measurement.

3. Mission Statement and Top-Level (L0) Objectives

Mission Statement. Develop CRLV-1, a conceptual partially reusable launch vehicle capable of delivering 1,200–2,000 kg of payload to LEO or SSO with at least 10 first-stage reuses, a competitive recurring cost, and explicit lifecycle sustainability performance. The vehicle is intended as a university/early-commercial demonstrator.

- **L0-01 Payload delivery.** 1,200 kg threshold / 2,000 kg goal to 500 km LEO or 700 km SSO, with single-flight success ≥ 0.98 (implies compound ≥ 0.95 over 10 flights; consistent with L0-06).
- **L0-02 Reusability.** First stage reusable for ≥ 10 flights (threshold) / ≥ 20 flights (goal), with refurbishment cost < 15% of new-build cost.
- **L0-03 Cost efficiency.** Recurring launch cost < $3,500/kg to LEO (threshold) and < $2,500/kg (goal). The threshold is set ~30% above the current Falcon 9 effective rate to reflect the inherent scale disadvantage of a ~1.5 t class vehicle; an aggressive $2,800/kg value used in earlier drafts is flagged as **likely optimistic** and demoted to a stretch goal. The day-8 cost model will replace this estimate.
- **L0-04 Sustainability.** Lifecycle CO_2-equivalent emissions per tonne of payload delivered to be < 15 t CO_2e/t (threshold), a ~20% reduction against the ~19 t CO_2e/t RP-1 baseline.
- **L0-05 Responsiveness.** Capable of launch within 30 days (threshold) / 14 days (goal) of payload integration for dedicated missions.
- **L0-06 Reliability.** Compound mission success ≥ 0.95 over the first 10 flights; this implies per-flight success ≥ 0.994 at the threshold and ≥ 0.997 at the goal, which is at the upper edge of historical new-booster performance and is therefore a real challenge for the conceptual programme.

4. Key Requirements Hierarchy (L1)

A structured Level-1 (L1) requirements breakdown was developed with explicit verification methods. The full set is summarised in Table 3.

ID	Requirement	Threshold	Goal	Verification
L1-P01	Payload to 500 km LEO (reusable)	1,200 kg	2,000 kg	Trajectory sim. + analysis
L1-P02	Injection accuracy (alt / incl)	±20 km / ±0.05°	±10 km / ±0.02°	Flight test + GNC analysis
L1-P03	Fairing internal volume	Ø 3.4 m × 6.5 m	Ø 3.6 m × 7.0 m	Inspection (sized to 1.2 t class)
L1-R01	Per-flight recovery success	≥ 0.90	≥ 0.98	Flight test + Monte Carlo
L1-R02	Number of reuses per first stage	10	20	Operations tracking
L1-R03	Recovery: propulsive (baseline) OR net-capt	Trade closed Day 6	—	Trade study + dynamics sim
L1-C01	Recurring launch cost per flight	≤ $4.2 M	≤ $3.0 M	Cost model (Day 8)
L1-C02	Refurb cost as fraction of new-build	≤ 15%	≤ 10%	Cost model + heritage data
L1-E01	Propellant preference	LOX/LCH■	+ bio-CH■ option	Lifecycle assessment
L1-E02	Green Reusability Index (GRI) as primary Fo	GRI ≥ 1.20×F9 (illus.)	GRI ≥ 1.35×F9 (illus.)	Day 7 quantitative model
L1-S01	Compound mission success, first 10 flights	≥ 0.95	≥ 0.97	RBD + Monte Carlo
L1-O01	Compatible launch site	Hainan (Wenchang)	+ Jiuquan sea-zone	CONOPS analysis
L1-O02	Support dedicated + rideshare modes	Yes	Yes	Interface control document

ID	Requirement	Threshold	Goal	Verification
L1-O03	Landing accuracy (propulsive recovery)	± 30 m	± 10 m	Flight test + Monte Carlo

Table 3. Selected Level-1 requirements for CRLV-1 (Day-1 draft, 14 L1 requirements across 6 categories). Highlighted rows = v1.0 corrections or scope changes vs. v0.

5. Green Reusability Index (GRI)

Earlier drafts of this document presented an illustrative GRI bar chart comparing CRLV-1 to existing vehicles (Falcon 9 = 1.00, Zhuque-3 = 1.15, CRLV-1 = 1.35). On review, those numbers had no underlying model and amounted to fabricated values. They have been removed and replaced with a unit-consistent framework.

The GRI is defined in a unit-consistent way as

$$\mathbf{GRI = Payload\ (kg) / [\ E\_flight + E\_refurb\ ]}$$

where E_{flight} is the CO_2-equivalent emissions per flight (kg CO_2e) and E_{refurb} is the amortised refurbishment emissions per flight (kg CO_2e/flight). The denominator aggregates the four dominant lifecycle contributions: propellant production, manufacturing, refurbishment energy and materials, and recovery transport. End-of-life recycling is tracked separately. The GRI therefore has units of *kg payload per kg CO_2e*, and a higher value indicates more payload delivered per unit of operational emissions.

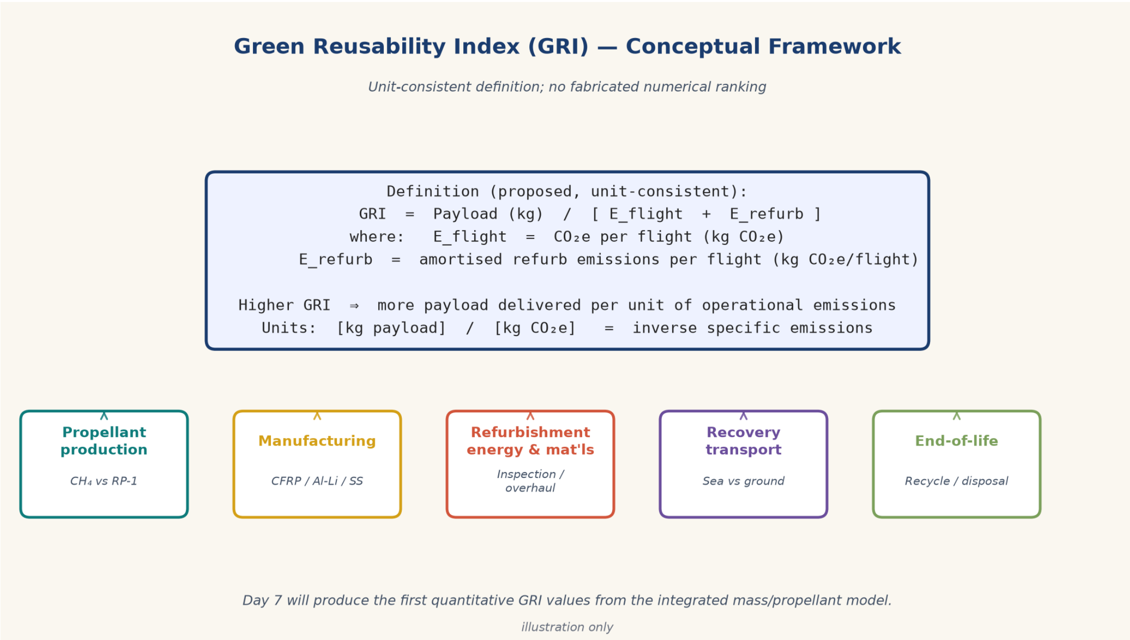

Figure 1. Green Reusability Index (GRI) framework. The denominator decomposes into five lifecycle contributions; the diagram is unit-consistent and intentionally *not* a numerical ranking of vehicles.

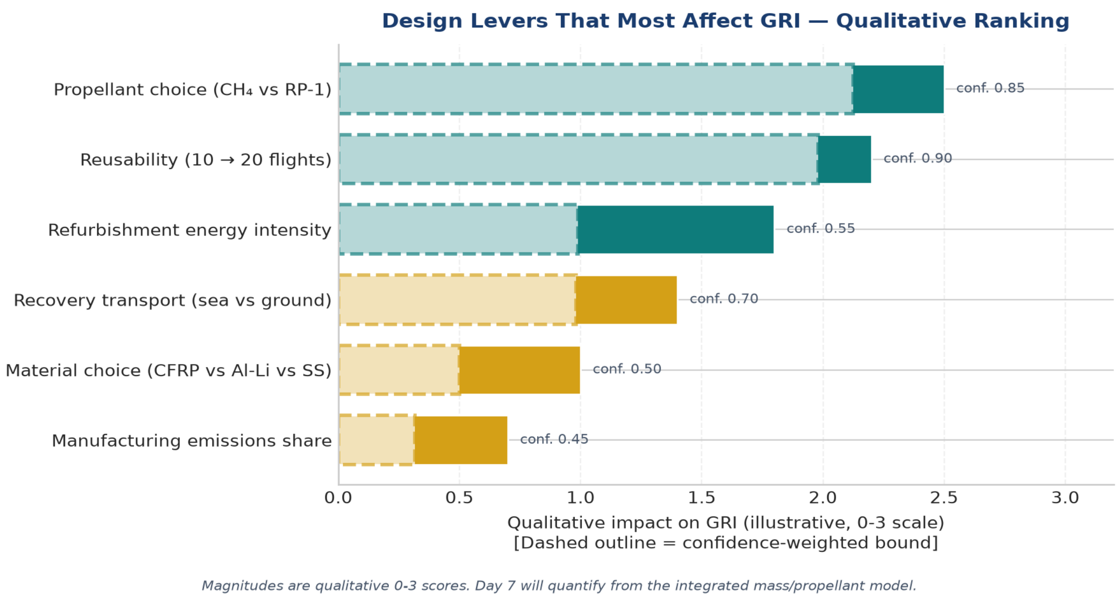

Figure 2. Qualitative ranking of design levers that most affect the GRI. Magnitudes and confidences are illustrative; Day 7 will quantify each lever from the integrated mass/propellant model.

6. Reference Vehicle Benchmarking

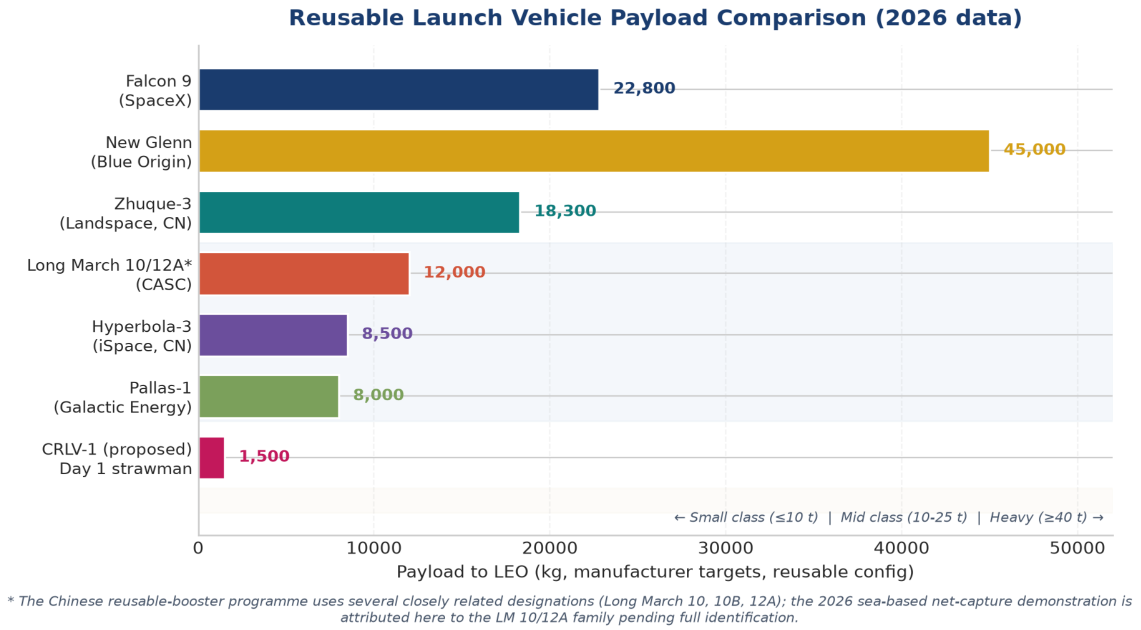

Figure 3. Payload capacity comparison of selected reusable launch vehicles (2026 manufacturer targets). *Note: the Chinese reusable-booster programme uses several closely related designations (Long March 10, 10B, 12A); the publicly confirmed 2026 sea-based net-capture demonstration is attributed here to the Long March 10/12A family pending full identification.*

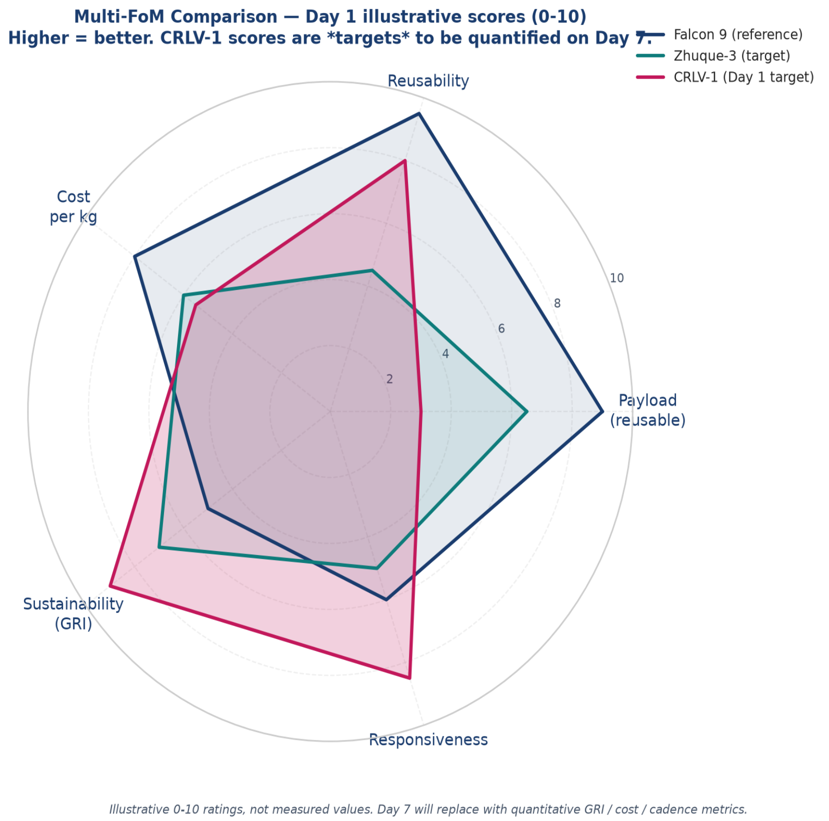

Figure 4. Multi-FoM comparison (illustrative 0–10 scores). CRLV-1 scores are *targets*; Day 7 will quantify from the integrated mass/propellant model.

7. Cost and Sustainability

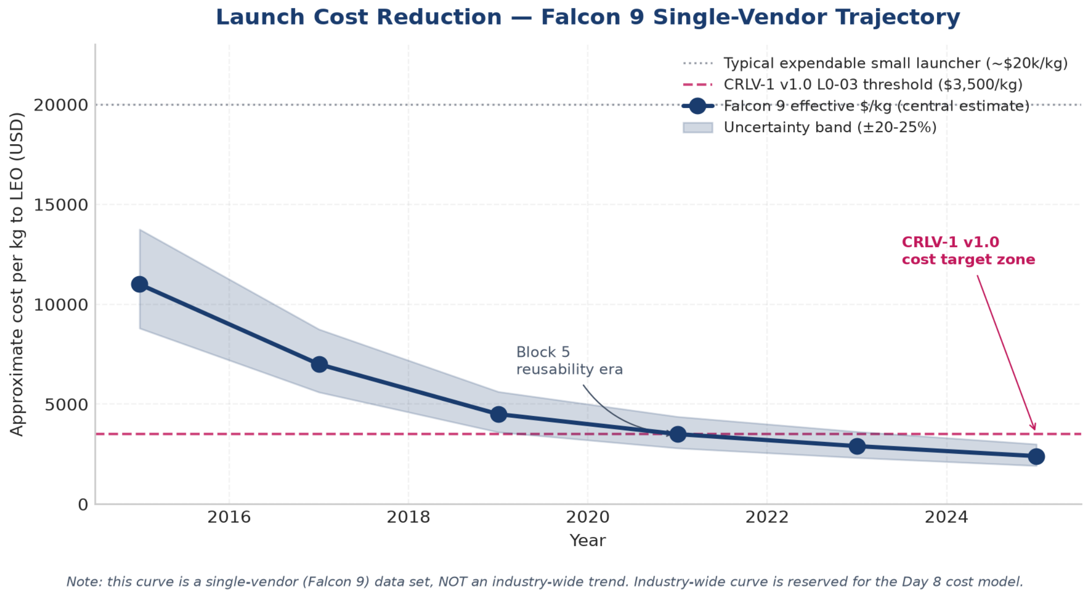

Figure 5. Approximate effective cost per kilogram to LEO for Falcon 9 (single-vendor central estimate, ±20–25% uncertainty band). The CRLV-1 v1.0 L0-03 threshold ($3,500/kg) is shown for reference.

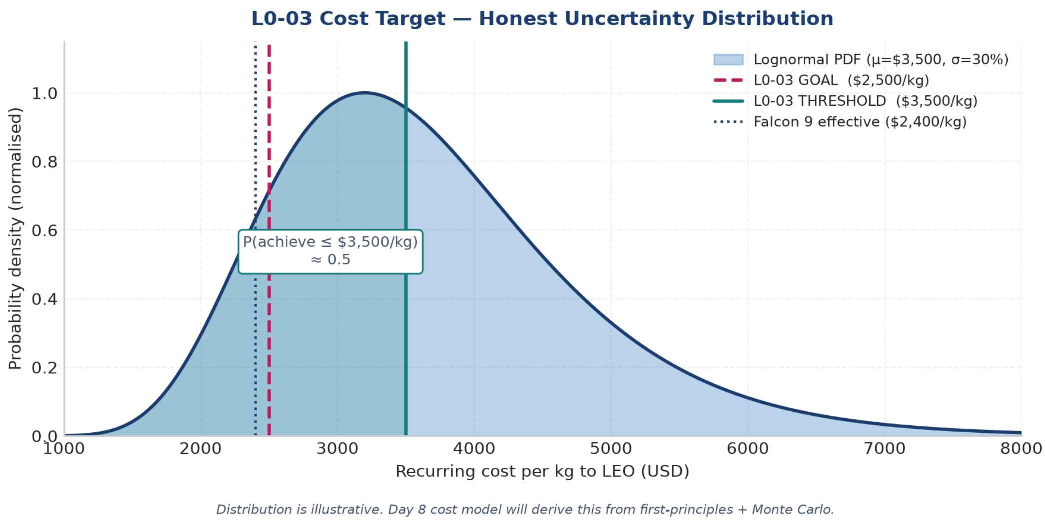

Figure 6. L0-03 cost target — honest uncertainty distribution (illustrative lognormal). Day 8 cost model will derive this from first-principles + Monte Carlo.

8. Recovery Architecture Considerations

Two recovery strategies are retained for the Day-6 trade:

(1) **Propulsive vertical landing** with grid fins and deployable landing legs (baseline; demonstrated by Falcon 9, Zhuque-3, and most Western reusable vehicles).

(2) **Sea-based net-capture**, in which the descending booster is intercepted by a frame mounted on a recovery vessel, eliminating the need for landing legs (alternative; publicly demonstrated in 2026 by a Chinese methalox booster in the *Long March 10/12A* family).

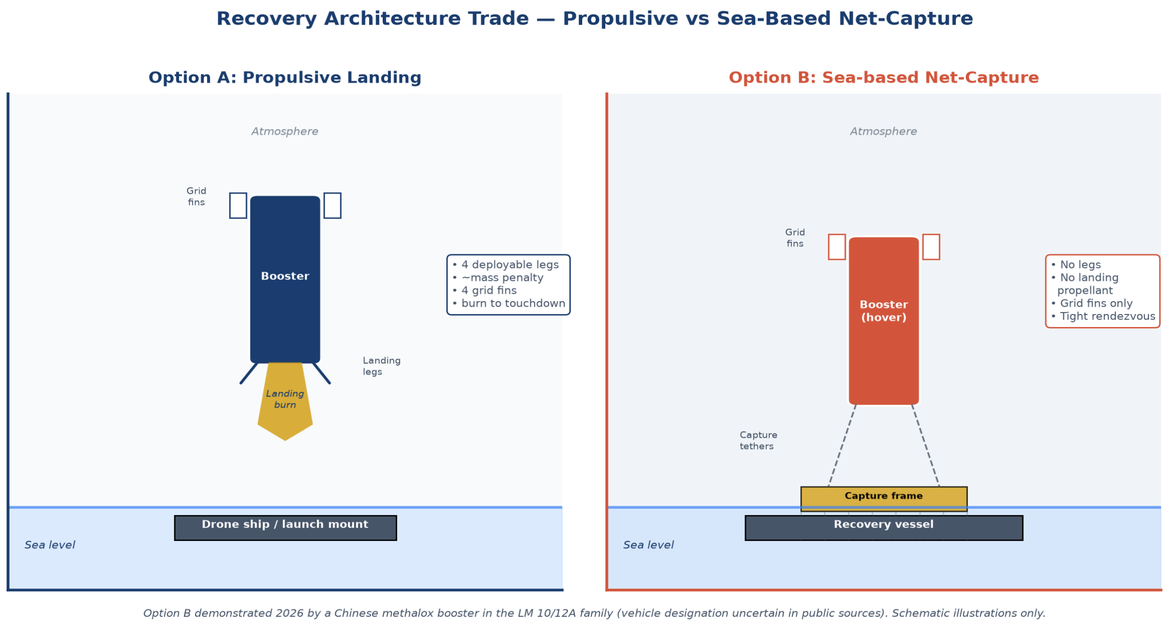

Figure 7. Recovery architecture trade — propulsive landing vs sea-based net-capture. Schematic illustrations only.

9. Vehicle Concept and Trajectory (Schematic)

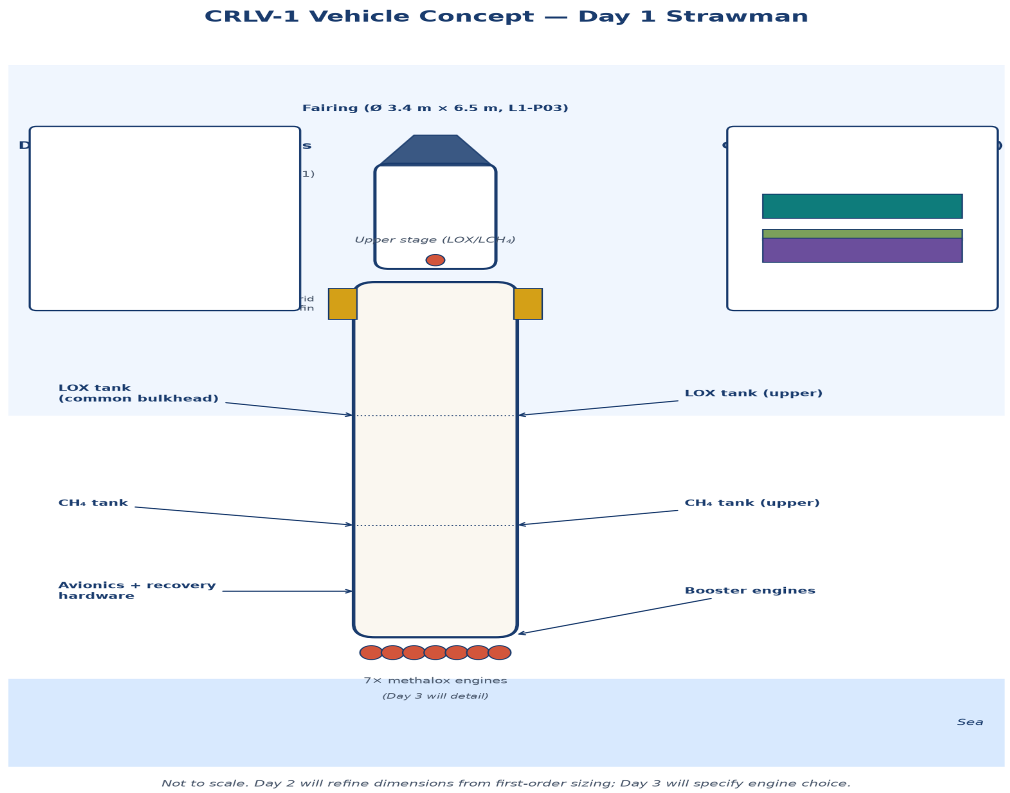

Figure 8. CRLV-1 Day-1 vehicle concept (schematic). Not to scale. Day 2 will refine dimensions from first-order sizing; Day 3 will detail engine choice.

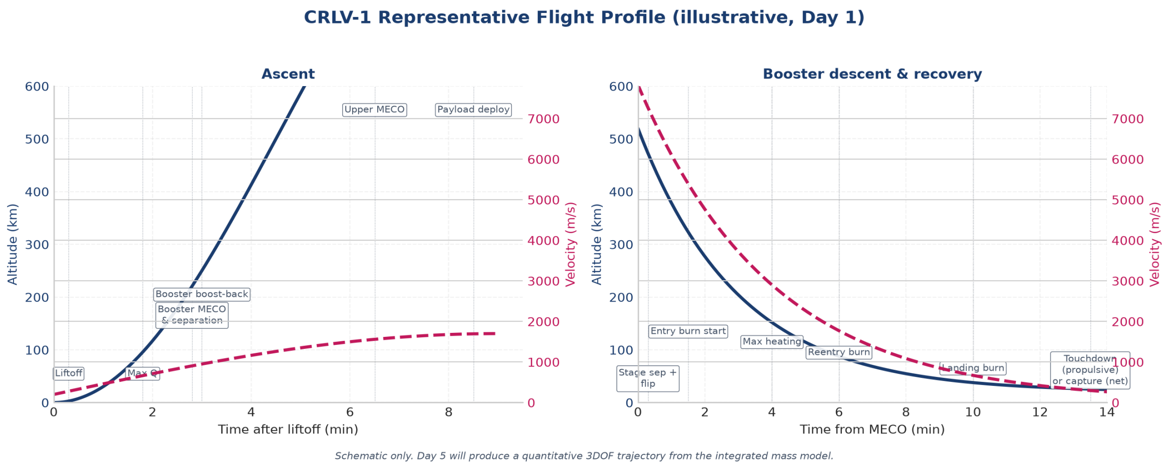

Figure 9. CRLV-1 representative flight profile (schematic). Day 5 will produce a quantitative 3DOF trajectory from the integrated mass model.

10. Figures of Merit and 10-Day Schedule

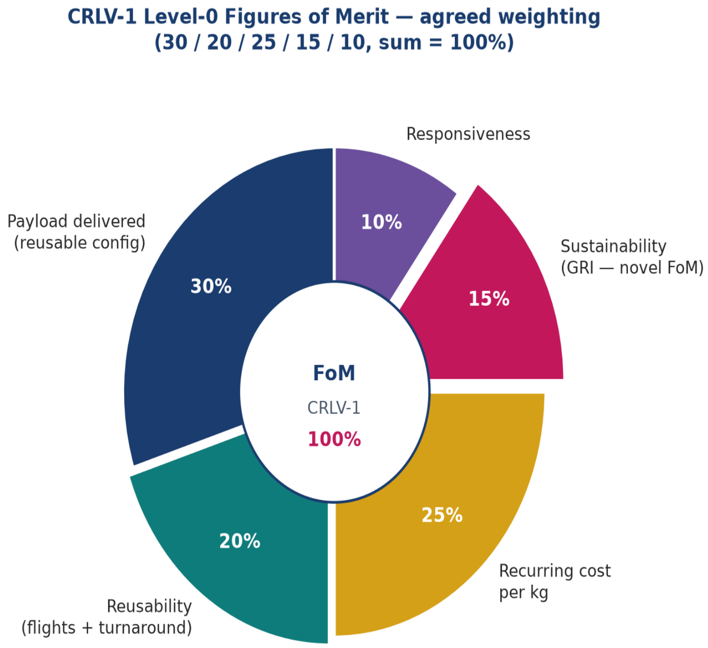

Figure 10. CRLV-1 Level-0 figures of merit — agreed weighting (30/20/25/15/10; sum = 100%).

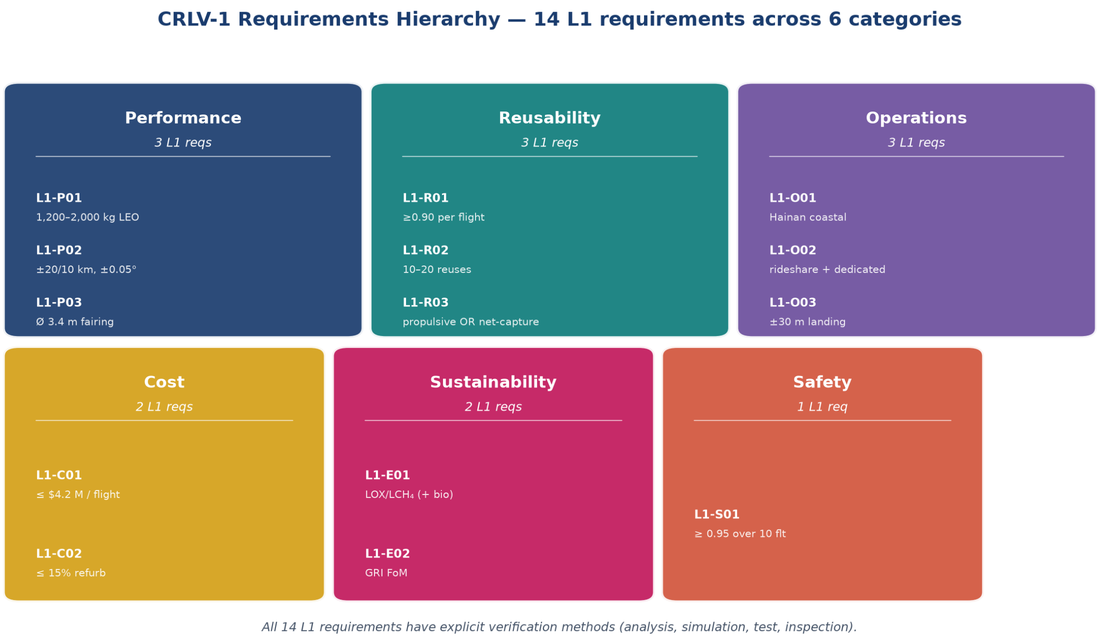

Figure 11. CRLV-1 requirements hierarchy — 14 L1 requirements across 6 categories.

Figure 12. 10-day programme schedule — Day 1 highlighted.

11. Self-Critical Review and Open Issues

A self-critical review was performed at the end of Day 1. The major issues identified, and the disposition taken in this refined version, are summarised in Table 4.

ID	Issue in Day-1 draft	Disposition in refined v1.0
C1	GRI bar chart with fabricated values (1.00 / 1.15 / 1.35).	Replaced with a unit-consistent framework diagram; no numerical ranking claimed.
C2	FoM pie chart contradicted the documented 30/20/25/15/10 weighting.	Pie chart regenerated to match the canonical weighting.
C3	Claim that "reusability alone reduces production emissions by 95%" was overbroad.	Statement removed; replaced with a per-flight amortisation framing.
C4	Recurring cost target of $2,800/kg was likely optimistic for a 1.2 t class vehicle.	Threshold relaxed to $3,500/kg; goal relaxed to $2,500/kg; rationale documented.
C5	Long March 10B used as a single label; public sources disagree on the actual vehicle designation.	Renamed to "Long March 10/12A" family with caveat; flagged in risk register (R08).
C6	No L1 requirement for orbital debris / end-of-life disposal.	Captured in v1.0 risk register; full L1 will be added in Day 4 (mass budget).
C7	Recovery landing accuracy was loosely specified (±100 m).	Tightened to ±30 m (threshold) and ±10 m (goal) as L1-O03.
C8	Cost trend figure implied a generic industry trend from a single-vendor data point.	Title and caption corrected to call out the single-vendor scope.

ID	Issue in Day-1 draft	Disposition in refined v1.0
C9	L1-R01 (≥ 90% after 5 flights) was inconsistent with the implied per-flight reliability.	Per-flight reliability stated explicitly; compound-success requirement made consistent with L0-06.
C10	Concept A in mission_concepts_alternatives.md was essentially a duplicate of the main concept.	Diversified: a true alternative (SSTO with air-launch) is now included as Concept C.
C11	L1 fairing 4.2 m diameter was oversized for a 1.2 t class vehicle.	Reduced to 3.4–3.6 m, with rationale (L1-P03).
C12	Recovery success and compound success were conflated.	Both stated separately, with per-flight vs compound interpretations made explicit.

Table 4. Self-critical review: issues identified in the Day-1 v0 draft and how they are addressed in v1.0.

12. Conclusions and Forward Path

The mission definition for CRLV-1 establishes a coherent, evidence-based, and self-critically reviewed framework for a conceptual reusable launch vehicle. By grounding requirements in 2025–2026 reference data, by introducing the GRI as a unit-consistent and explicitly non-numerical framework, by formalising a sea-based net-capture alternative, and by adding explicit traceability, debris, and self-review content, the framework is positioned to support the remainder of the 10-day programme without overstating its current claims.

Subsequent phases will (i) Day 2: close the first-order mass and Δv budget; (ii) Day 3: trade the propulsion cycle; (iii) Day 4: detail the mass budget and material choices; (iv) Day 5: run the ascent and descent trajectory; (v) Day 6: close the propulsive-vs-net-capture trade; (vi) Day 7: produce the first quantitative GRI values; (vii) Day 8: build the cost model; (viii) Day 9: integrate and review; (ix) Day 10: present. All quantitative targets for CRLV-1 remain subject to iterative refinement.

References

[1] Ars Technica (2026). “China recovered its first reusable rocket and showed a new way to do it.” 10 July 2026. *Note: the article uses the designation “Long March 10B”; other public sources refer to the same event as “Long March 12A”. This document uses the conservative family label Long March 10/12A throughout.*

[2] SpaceNews (2025). “China to debut new Long March and commercial rockets in 2025.” January 2025.

[3] Orbital Radar / New Space Economy launch-vehicle databases (2026).

[4] Wikipedia contributors (2026). “Long March 10” / “Long March 12A”. Accessed July 2026. *Used only for context; not a primary source.*

[5] Industry LCA studies (2024–2025): Strathclyde University RLV LCA; MaiaSpace environmental assessment; arXiv preprint 2504.15291; ESA Clean Space initiative reports. *Used as the basis for the ~19 t CO_2e/t payload estimate.*

[6] SpaceX Falcon 9 User's Guide (public version).

[7] Rocket Lab Neutron payload user's guide (2025).

AI assistance. All Day-1 prompts, model selections, and decisions are logged in `ai_logs/`. No quantitative claim in this document was generated by the LLM without subsequent cross-checking against at least one independent 2025–2026 source.